

On NextG Open RAN as a Sensing Infrastructure

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Abstract

While communication and sensing systems have historically evolved independently, there are several compelling reasons for their integrated operation, including reduction in deployment and operational costs. We consider different ways to integrate communication and sensing, and argue that joint communication and sensing (JCAS) paradigm stands out clearly as the preferred alternative, that too from a network-centric perspective. Given the above, we make the case for turning the NextG radio network infrastructure for enabling sensing applications, leveraging the trend towards Open RAN adoption in NextG networks. We outline NextG Open RAN enabled sensing applications spanning both spectrum sensing and wireless signal sensing. We discuss several research challenges that need to be addressed to realize NextG Open RAN based sensing. We consider device positioning as a representative sensing use case in this setting and highlight some challenges for robust positioning through analysis of measurements from a carrier-grade indoor 5G Open RAN testbed.

CCS Concepts

• **Networks** → **Mobile networks**; • **Hardware** → **Wireless integrated network sensors**.

Keywords

Joint Communication and Sensing (JCAS); NextG Networks; Open RAN; Wireless Sensing and Positioning

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1 Introduction

Communication and sensing systems have been developed with distinct objectives. Communication systems, such as mobile cellular networks, use radio waves to connect mobile devices to a network of base stations, facilitating voice, data, and messaging services across wide areas. On the other hand, sensing systems like RADAR employ radio waves to detect, measure, and track objects' distance, speed, and direction by analyzing the reflections of emitted signals. As such, communication and sensing systems have traditionally evolved along parallel paths. But considerable efforts are being been made to integrate or enable cooperation between them [11, 16, 17, 35]. We believe there are key benefits from their integration. First, these systems share hardware and software commonalities, using similar components like transmitters, receivers, and antennas, as well as signal processing techniques such as matched filtering and FFT. Second, modern applications (e.g., autonomous vehicles, smart cities, industrial IoT, healthcare) need both communication and sensing. For instance, autonomous vehicles use RADAR and LIDAR for navigation while relying on communication systems for data exchange with the infrastructure for enhancing safety and operational efficiency. Third and crucially, their integration offers significant benefits by reducing deployment and operational costs through the use of existing hardware, lowering power consumption, and better spectrum use.

As illustrated in Fig. 1, the current approaches that integrate communication and sensing can be broadly classified into three categories [35]:

(1) **Coexisting Systems** – Separate signals are used for communication and sensing, either by employing frequency or time division to simplify data processing, or through more complex techniques to share channel resources simultaneously.
(2) **Passive Sensing** – This method involves analyzing received signals through specialized receiver designs, operating independently of the communication system.
(3) **Joint Communication and Sensing (JCAS)** – This approach utilizes a joint signal to achieve both communication and sensing objectives, optimizing waveform, system, and network design for integrated performance.

Among the above approaches, the JCAS solution stands out for its high spectrum efficiency and shared (hardware and software) resources for communication and sensing. Depending on the system architecture, JCAS can be further categorized into mono-static, downlink bi-static, and uplink bi-static configurations, as elaborated in §2.

In this paper, we bring together arguments to make the case for leveraging the 5G and beyond (NextG) radio network infrastructure based on Open Radio Access Network (RAN) [22] as a sensing infrastructure, following the JCAS paradigm and using the uplink bi-static configuration outlined above. The foremost reason supporting this case is the ubiquitous coverage and continual evolution of cellular networks. With over 90% of the global population served by mobile broadband and more than 20 million base stations worldwide [12], leveraging this global scale radio network infrastructure for sensing enables ubiquitous sensing and applications across industries [8, 25]. Furthermore, 5G is the ideal choice for integration, offering scalability and the capability to support high bandwidth, low latency, and massive connectivity for advanced sensing applications.

The emergence of Open RAN and its likely adoption in NextG networks is a significant and timely contributing factor to this case.



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Unlike in traditional RANs where the RAN data and its use is hidden inside proprietary and monolithic RAN equipment from a few vendors, Open RAN through disaggregation of RAN components changes the paradigm of RAN operation to be data driven. All the functionality realized previously inside the RAN components can now be realized as an "App" over the RAN Intelligent Controllers (RICs) using the data from the RAN. Furthermore, each of those Apps can be innovated leveraging the power of AI. This paradigm shift in the RAN architecture not only allows rethinking RAN operation aimed at resource allocation and network management for improved efficiencies and greater automation but also opens the door for imagining new types of data-driven RAN Apps that are not limited to improving RAN operations. As a result, we can envision a suite of sensing related Apps focusing on sensing of various kinds (including positioning) that are enabled by data from the underlying RAN infrastructure.

As an example, in the work of Smith et al. [27], the data-driven and App-oriented operation of Open RAN is utilized to detect channel vacancies using machine learning algorithms (e.g., CNN) without the need for additional equipment or waveform modifications, effectively harnessing unused wireless spectrum resources. This example illustrates how the NextG Open RAN architecture supports sensing applications. However, the potential of JCAS in this setting extends far beyond spectrum management. Wi-Fi sensing applications offer valuable insights, as both Wi-Fi and mobile networks use OFDM symbols and report Channel State Information (CSI) for communication [34]. By leveraging the ease of flexibly accessing RAN data with Open RAN and integrating machine learning techniques, it is possible to not only replicate sensing tasks which are typically achieved with commodity Wi-Fi devices [18] (e.g., positioning, traffic monitoring, motion recognition) over the NextG network infrastructure, but also to achieve superior performance and broader coverage.

In the rest of the paper, we elaborate our case for using NextG Open RAN based infrastructure for sensing by:

- discussing the advantages and limitations of current solutions for integrating communication and sensing, and explaining the feasibility of achieving JCAS in this setting.
- categorizing sensing applications into spectrum sensing and wireless sensing, and outlining the key challenges and research questions to realize JCAS over NextG Open RAN.
- using device localization as a representative sensing use case based on 5G reference signal measurements from an indoor carrier-grade 5G Open RAN testbed [6]. This case study demonstrates how communication reference signals can be leveraged to enable sensing functionalities as well as highlights the associated challenges.

2 Background

In this section, we first provide an overview of Open RAN and associated O-RAN standards as we believe it is a key enabler for integrating communication and sensing in NextG networks. Then we explain the three categories of integrating communication and sensing. While these categories can support similar sensing tasks, their performance and implementation complexity vary. Then, we further introduce the JCAS system from a network-centric perspective. Compared to device-centric sensing (that effectively is akin to

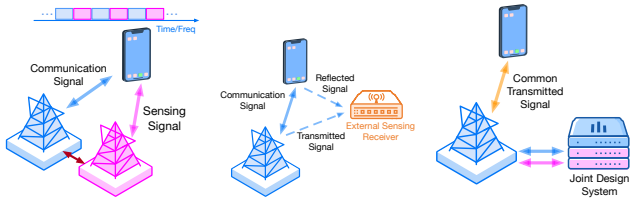


Figure 1: Different paradigms for integrated communication and sensing systems.

passive sensing), network-centric systems offer several advantages, including greater computational power, stable networks, and easier multi-node cooperation, thereby making them the preferred choice for JCAS.

Open RAN and O-RAN Open RAN [28] refers to the industry-wide initiative to dis-aggregate radio base station hardware and software, enabling open interfaces between subsystems to reduce the vendor lock-in and promote a more competitive RAN ecosystem. Within this context, the O-RAN Alliance [1] plays a key role by defining open RAN specifications, releasing open-source software, and supporting the integration and testing of diverse implementations. Central to the O-RAN architecture is the functional decomposition of radio base stations into modular components – Radio Unit (RU), Distributed Unit (DU), and Centralized Unit (CU) – connected via standardized "open" interfaces, such as the 7.2x fronthaul, E2, A1, and O1 interfaces, ensuring vendor interoperability and seamless orchestration.

The O-RAN framework [22] introduces RAN Intelligent Controllers (RICs) for programmable control and machine learning-driven optimization, with near-real-time (near-RT) RICs enabling low-latency xApps and non-real-time (non-RT) RICs handling policy and model training through rApps. The alliance also actively collaborates with organizations like the Linux Foundation and TIP, providing open-source tools and fostering a unified development ecosystem. By emphasizing innovation, interoperability, and open-source contributions, the O-RAN Alliance accelerates the adoption of disaggregated RAN systems while advancing the broader goals of the open RAN movement.

Communication and Sensing Integration. As previously discussed, current approaches that combine communication and sensing can generally be categorized into Coexisting Systems, Passive Sensing Systems, and Joint Communication and Sensing (JCAS) Systems, whose structures are illustrated in Fig. 1.

In Coexisting Systems [11], as shown in Fig. 1a, hardware and software may be partially shared, but the communication and sensing signals are typically separated. The signals can be divided by time, frequency, code, or polarization (e.g., [11, 24]). Some coexisting systems share resources such as time and frequency, which require more complex and specially designed signal processing algorithms to manage interference issues [15, 19].

For Passive Sensing Systems (e.g., [3]), no additional signal is transmitted specifically for sensing. Instead, an external sensing receiver is added. The advantage of this approach is that no modifications are required to the existing communication system. However, the sensing capability is limited by the external device and the

complexity of the communication signals. Without direct support from the communication side, some cases like sensing multiple transmitters or coherence sensing could become more challenging.

In JCAS systems [16, 17], the transmitted signal is designed to serve both communication and sensing purposes, with the system being jointly optimized. This structure offers the best spectral and temporal efficiency among the three approaches. So, we focus on JCAS systems from here on.

Network-Centric JCAS System. Radar systems can be categorized as mono-static or bi-static, depending on the spatial arrangement of the transmitter and receiver. Mono-static systems have collocated transmitters and receivers, while bi-static systems use spatially separated components (Figure 2). More complex configurations with multiple transmitters and receivers are known as multi-static [26]. Similarly, JCAS sensing capabilities can be classified into mono-static and bi-static [23]. In mono-static JCAS (Figure 3), the same base station (gNB in 5G) transmits and receives the signals reflected off the target [36]. Bi-static JCAS can be split into downlink and uplink types: the downlink system receives signals transmitted by the gNB, while the uplink system captures signals from the UE [23].

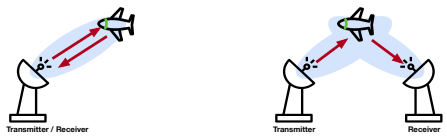
Mono-static and downlink bi-static systems, which use signals transmitted by the gNB, provide stronger signal power and higher signal-to-noise ratio (SNR). However, receiving signals from the gNB itself or nearby base stations can interfere with the priority of data transmission. In contrast, the uplink bi-static system aligns with the standard gNB-UE communication model and requires minimal customization for implementation [23]. Thus, our use case is based on this configuration.

Signals for Sensing. In general mobile networks, as illustrated in Fig. 3, data frames consist of synchronization signals (e.g, PSS, SSS), reference signals (e.g, DMRS, CSI-RS, PT-RS, RIM-RS, SRS, PRS [2]), and data payloads. Synchronization signals enable time and frequency alignment between the gNB and UE, reference signals assist with channel estimation for reliable communication, and data payloads carry user data or control information.

Reference signals serve various functions, primarily supporting communication quality. For example, CSI-RS is used for beam management and CSI acquisition, PTRS for phase tracking, and DMRS for demodulation. However, to enable positioning services starting with Release 16, PRS was introduced in the downlink to enhance positioning accuracy, offering higher density per PRB and configurable bandwidth. In the uplink, SRS is used for positioning, with improvements in Release 16 through staged resource element mapping and configurable numbers of OFDM symbols [14]. Furthermore, combining existing signals can enhance positioning and sensing performance by enabling continuous time-frequency resource coverage, as individual reference signals typically follow a comb-like distribution. For instance, collaboratively using DMRS, PRS, and CSI-RS ensures continuous distribution across both time and frequency domains, resulting in higher sensing accuracy and extending the maximum unambiguous distance and velocity [30].

3 NextG Open RAN Enabled Sensing Applications

Here we outline potential JCAS applications, broadly categorized into spectrum sensing and wireless signal sensing.



(a) Mono-static (b) Bi-static

Figure 2: RADAR Systems

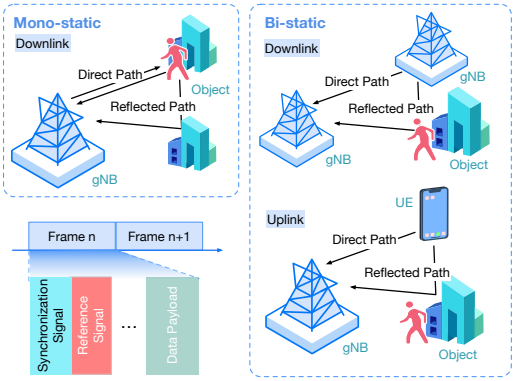


Figure 3: Network-Centric JCAS System Variants

Spectrum Sensing. The main goal of spectrum sensing is to detect energy within a target frequency band to determine spectrum occupancy [5]. This capability enables various applications. First, it facilitates interference detection by identifying the source and nature of external interference, which improves signal quality, reduces errors, and lowers latency, thereby enhancing communication reliability. Preemptively addressing malicious spectrum use also strengthens network security by mitigating threats like rogue base stations and signal hijacking.

Additionally, spectrum sensing supports dynamic channel selection, flexible spectrum sharing, and allocation [4]. For example, [27] proposes enabling secondary users to access underutilized government or military bands when primary users are inactive. Instead of adding extra sensors, this functionality can be implemented by feeding data collected by the RAN into the intelligent network manager within the RIC. The fine-grained control offered by Open-RAN allows for better performance using advanced techniques like physical resource block (PRB) blanking, enabling interference mitigation and real-time closed-loop control.

Wireless Signal Sensing. Wireless signal sensing uses the propagation channel to extract environmental information or detect objects [7]. These applications adapt radar sensing for mobile scenarios, mirroring OFDM-based systems like Wi-Fi [18]. A fundamental application is positioning, already standardized for connected 5G devices in 3GPP technical specifications. Accurate location data reduces beamforming delays and improves channel quality, enhancing communication. Positioning also serves as a foundation for advanced systems like smart cities and vehicular networks (V2X), where seamless collaboration is critical.

Beyond connected devices, the signals exchanged between gNB and UEs can serve as sources of situational data, conveying information about objects along the signal path, including their shape, speed, and even the materials they are made of [21]. The capability to perform device-free positioning, sensing, and detection has

been identified as a key open research question in the vision for 6G [25]. This functionality will enable innovative applications such as human activity recognition, simultaneous localization and mapping (SLAM) and object tracking, while also supporting situational awareness services like urban traffic monitoring, smart building occupancy detection, and environmental sensing for emergency response.

4 Research Challenges

In this section, we discuss four potential challenges and research questions in the implementation and development of network-centric JCAS systems.

Synchronization and Signal Processing. As previously mentioned, all bi-static JCAS systems share the benefit of obtaining sensing data without altering communication processes. However, the lack of clock synchronization remains an inherent challenge in sensing, such as time-of-flight (ToF) and Doppler estimation. This issue arises because each device's clock is generated by its local oscillator (LO), leading to various interferences in the received signals, such as Carrier Frequency Offset (CFO), Timing Offset (TO), and Phase Offsets (PO) [31]. CFO is caused by discrepancies in oscillator-generated frequencies, leading to variations in carrier frequencies. TO, such as Symbol Time Offset (STO), results from unsynchronized clocks and includes timing errors due to signal propagation delays. Additionally, phase errors can arise from various sources like inaccuracies in the LO or thermal noise, and these errors may fluctuate rapidly from symbol to symbol.

Beyond clock synchronization related errors, environmental interferences also affect sensing accuracy. Issues like multipath interference, hardware differences, and the limitations of sensing precision due to frequency and bandwidth constraints are common to all bi-static sensing systems. While there is extensive related research on these challenges (e.g, Wi-Fi Sensing [29, 33]), further practical validation is necessary to apply these solutions to real-world 5G uplink systems.

Data Compression. One key challenge is the compression of data collected by the RU (Radio Unit) for use at the higher layer RIC and Apps. When RUs act as sensors, the data they gather must be processed and communicated over the mid-haul/backhaul network to where RICs and Apps are hosted. However, transmitting all in-phase and quadrature (IQ) data results in an overwhelming volume, consuming significant bandwidth. For instance, with four antennas transmitting one demodulation reference signal (DMRS) symbol per slot and a 100MHz bandwidth at a 30 kHz subcarrier spacing, a single RU generates 213.33 MB of data per second. The burden of transmitting this data increases linearly with the number of RUs, leading to high bandwidth consumption and limiting real-time application performance.

Data compression faces several challenges [32]. First, communicating sensing related data (e.g., IQ samples) from RUs via DUs to higher layer RICs is constrained by limited processing capabilities at far-edge sites, making it difficult to run complex compression algorithms, and may also disrupt the real-time processing in the DU. Further, bandwidth consumption associated with communicating this data contends with production RAN (control/data plane) traffic over the mid-haul/backhaul links. Additionally, the compression rate must be dynamically adjustable, as the ideal rate varies with different channel conditions and applications. Establishing a dynamic feedback mechanism for adjusting compression is crucial. Lastly, existing compression algorithms are designed for communication systems, and different applications may require different algorithms. For example, in interference detection through power spectrum density (PSD), simple filtering may suffice. In contrast, more detailed sensing tasks like positioning, tracking, or breathing detection may require richer data on wireless signal propagation, including amplitude and phase variations.

Communication Data Utilization. As previously discussed, communication centric JCAS systems aim to incorporate sensing functionalities by processing existing communication data without significantly altering the system architecture. One major challenge is effectively utilizing various physical channels or reference signals to enhance JCAS performance. For instance, combining multiple reference signals can improve temporal and frequency resolution. [30] proposed leveraging the differences in temporal and frequency distributions between PRS, CSI-RS, and DMRS to enable continuous sampling in both time and frequency.

Besides processing raw physical layer signals, another approach involves extracting sensing features from higher-level Key Performance Indicators (KPIs) or computed parameters. A typical application is fingerprinting based localization [13, 20], where KPIs such as Signal-to-Interference-plus-Noise Ratio (SINR), Reference Signal Received Power (RSRP), Reference Signal Received Quality (RSRQ), or Received Signal Strength Indicator (RSSI) are utilized. Some studies also consider uplink bandwidth (ULB), downlink bandwidth (DLB), and Physical Cell Identity (PCI). Experimental results have shown that using these KPIs can improve localization accuracy.

Multiple-gNB Collaboration. Network-centric sensing has several advantages over User Equipment (UE) based sensing, including better computational resources and superior signal quality. A significant advantage is the ability to receive data transmitted by a single UE from multiple gNBs/RUs, each with a known location and orientation. This multi-gNB setup effectively provides multiple observation points for the same target, enhancing application performance [10]. A basic example of this is triangulation-based localization. With only one observation point, the Angle of Arrival (AoA) alone cannot pinpoint the exact position. However, with two observation points, the intersection of the AoA lines can accurately determine the UE's location on a plane.

Another promising application is detection of user device context (indoor /outdoor, mobility, etc.), which is beneficial for communication and sensing tasks, such as predicting the user behavior based on movement patterns or optimizing resource allocation according to network load. The context information plays a vital role in confirming UE mobility, detecting environment changes, and understanding correlations between different UEs.

5 Device Localization Use Case

This section introduces our use case, which serves as the first step in the development of an Open-RAN based network-centric JSAC system. The application highlighted in this use case is positioning. It is chosen not only because it is one of the most valuable wireless sensing services, but also because it provides essential measurements for other applications, such as VEX. Our use case implements a positioning application based on uplink DMRS symbols and a

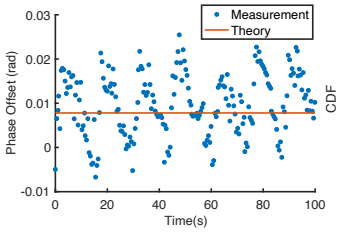


Figure 4: Phase Offset

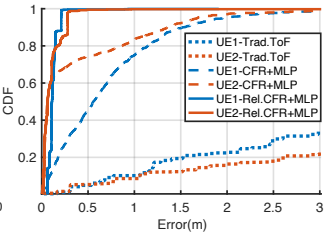


Figure 5: Localization Error

machine learning based position estimation using a multi-layer perceptron (MLP) based neural network.

Testbed and Measurements. We obtained DMRS measurements from a carrier-grade indoor 5G Open RAN testbed [6] with a spatially distributed set of devices (UEs), leveraging the Janus Open RAN telemetry system [9] integrated into the testbed. Our experiment setup involves one RU and one UE, with three different UEs included in the trials. The PUSCH operates with a subcarrier spacing of 30 kHz and uses 273 Resource Elements (REs) in total, with each RE corresponding to 12 subcarriers. The DMRS uses Mapping Type A, meaning that half of the total subcarriers are allocated for DMRS sequences, distributed non-continuously with one subcarrier spacing. Data sampling is performed once per slot for every frame, with one sample extracted every 50 frames. The RU is equipped with four antennas, and the Channel Frequency Response (CFR) is collected for each antenna element, which is then used as the feature for position estimation.

These settings are not specifically designed for sensing applications but reflect general communication configurations. Reference signal symbols are directly extracted from the physical layer channel for positioning, and additional data processing is performed on them. In other words, we consider this testbed suitable as a baseline for the JCAS system, enabling sensing without disrupting ongoing communication processes.

Evaluation. As an initial exploration, a simple Line-of-Sight (LOS) dataset is processed using a basic MLP model in our experiment. We believe this experiment not only serves as a baseline but also as a case study to better illustrate some of the research challenges discussed earlier. Data is collected in a hallway setting. The RU is fixed near the ceiling on one side of the hallway, while the UE starts directly below the RU and moves towards the opposite side, collecting data at 2-meter intervals, with the furthest distance being 18 meters, covering a total of 10 positions. We used two 5G smartphones of the same model (Oneplus Nord) to evaluate the device heterogeneity. The CFR estimated from DMRS is utilized as the feature, describing the signal propagation process.

Synchronization Issue and Phase Offset Effect: One of the key challenges is managing bi-static synchronization. Figure 4 shows the phase offset between adjacent subcarriers in the CFR when the UE is positioned 6 meters from the RU. The "Measurement" phase offsets are obtained by applying a linear fit to the measured CFR phase, while the "Theory" represents the theoretical phase offset, calculated using the formula $\Delta\phi = \frac{2\pi \cdot d\Delta f}{c}$, where c is the speed of light, $\Delta\phi$ is the phase difference, and Δf is the subcarrier spacing. Due to residual phase interference caused by synchronization issues and other factors, time-varying errors become a significant source of inaccuracy. Based on the formula, a phase difference of 0.0013 radians leads to a 1-meter error.

Table 1: Device Heterogeneity Test

	$UE1_{ID}$	$UE1_{OOD}$	$UE2_{ID}$	$UE2_{OOD}$
Error P50 (m)	0.11	7.20	0.07	4.14
Error P90 (m)	0.21	18.67	0.28	12.37

We aimed to explore whether machine learning techniques could mitigate the impact of phase offset. The positioning error was used as the evaluation metric, as shown in Fig. 5. Three different approaches were applied for positioning with two UE devices of the same model. The first approach is ToF estimation using the inverse fast Fourier transform (IFFT), referred to as "Trad. ToF." The second and third approaches utilize a MLP model but with different input formats. "CFR" represents the raw CFR input, which includes residual phase interference, while "Rel. CFR" represents the ratio of CFR values between two antennas, which reduces the interference caused by using the same local oscillator (LO) source [33].

As illustrated in Fig. 5, the 90th percentile of positioning accuracy using the raw CFR input is 1.52 meters and 1.35 meters, respectively. This is a significant improvement compared to the 90th percentile error of 22.6 meters with the traditional ToF method. However, the optimized Relative CFR vector performed even better, achieving 90th percentile errors of 0.21 meters and 0.28 meters. These results indicate that while machine learning can address some data processing challenges, combining it with traditional signal processing methods can yield even more accurate results.

Device Heterogeneity: We explored the impact of device heterogeneity on positioning accuracy by training the model separately on each device's data and testing it with both in-distribution (ID) and out-of-distribution (OOD) data. Here, ID data refers to data from the same UE used for training, while OOD data refers to data from a different UE. Both datasets used CFR_{rel} to reduce the impact of phase offset from the transmitter side. The results, shown in Table 1, reveal that even when the two UEs are of the same model, predicting with OOD data results in a significant error increase, with the 50th percentile error rising by at least 6.5 ×. This highlights the inherent differences between devices and suggests that addressing robustness and cross-device generalization is crucial for closing the gap to practical application.

6 Conclusions

We have analyzed the potential of Open RAN based NextG networks to integrate communication and sensing, arguing for the feasibility of achieving network-centric JCAS. We explored the advantages and feasibility of various methods for implementing JCAS systems and examine the potential of using existing communication signal types for sensing. Additionally, we have highlighted different types of sensing applications and discuss the research challenges that need to be addressed for enabling these applications. Finally, we use positioning as a use case to validate the feasibility of using DMRS reference signals as a signal source, combined with MLP based positioning estimation, based on measurements from a carrier-grade 5G Open RAN testbed. This practical application

not only demonstrates how existing communication reference signals can be leveraged to enable sensing functionalities but also highlights the associated challenges.

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